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Lecture 18 Tree-based Methods III: Bagging and Random Forest

STAT 441: Statistical Methods for Learning and Data Mining

Wenlu TANG

Department of Mathematical and Statistical Sciences
University of Alberta

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Outline

1. Bagging
2. Random Forests
3. Summary and Remark

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Bagging

- Bootstrap aggregation, or bagging, is a general-purpose procedure for reducing the variance of a statistical learning method;
- We introduce it here because it is particularly useful and frequently used in the context of decision trees.
- Recall that given a set of n independent observations $Z_1, \ldots, Z_n$, each with variance σ^2, the variance of the mean $\bar{Z}$ of the observations is given by σ^2/n.
- In other words, averaging a set of observations reduces variance. Of course, this is not practical because we generally do not have access to multiple training sets.

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Bagging

- Instead, we can bootstrap, by taking repeated samples from the (single) training data set.
- In this approach we generate B different bootstrapped training data sets.
- We then train our method on the b-th bootstrapped training set in order to get $\hat{f}^{*b}(x)$, the prediction at a point x.
- We then average all the predictions to obtain

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B}\hat{f}^{*b}(x).$$

- This is called bagging.

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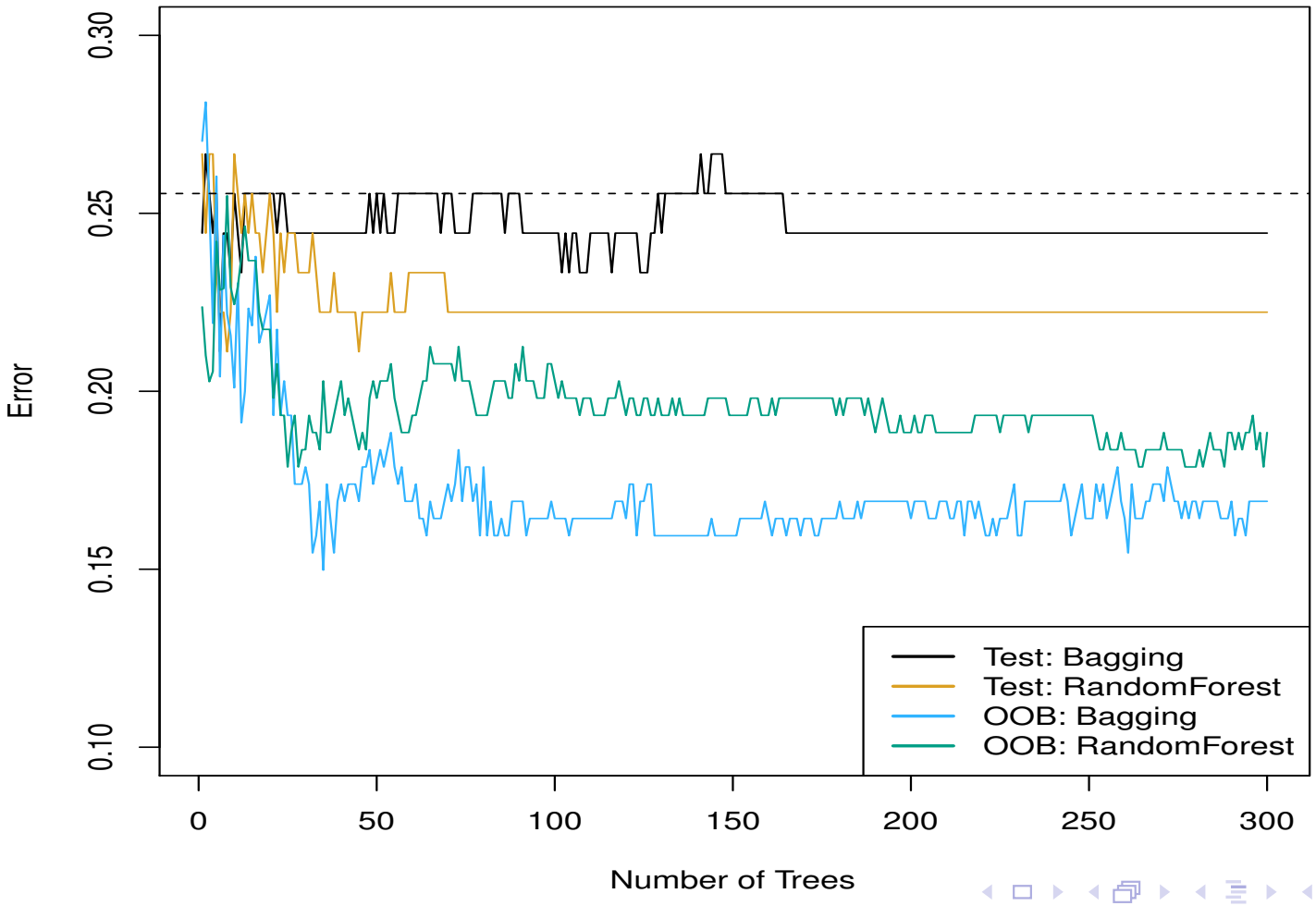
Bagging classification trees

- The above prescription applied to regression trees.
- For classification trees: for each test observation, we record the class predicted by each of the B trees, and take a majority vote: the overall prediction is the most commonly occurring class among the B predictions.

Bagging the Heart Data



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Details of previous figure

Bagging and random forest results for the Heart data.

- The test error (black and orange) is shown as a function of B, the number of bootstrapped training sets used.
- Random forests were applied with $m = \sqrt{p}$.
- The dashed line indicates the test error resulting from a single classification tree.
- The green and blue traces show the OOB error, which in this case is considerably lower.

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Out-of-Bag Error Estimation

- It turns out that there is a very straightforward way to estimate the test error of a bagged model.
- Recall that the key to bagging is that trees are repeatedly fit to bootstrapped subsets of the observations. One can show that on average, each bagged tree makes use of around two-thirds of the observations.
- The remaining one-third of the observations not used to fit a given bagged tree are referred to as the out-of-bag (OOB) observations.
- We can predict the response for the ith observation using each of the trees in which that observation was OOB. This will yield around $B/3$ predictions for the i-th observation, which we average.
- This estimate is essentially the LOO cross-validation error for bagging, if B is large.

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Extension: Random Forest

- How individual decisions trees are combined to make a random forest?
- Why random forests are so good at what they do.

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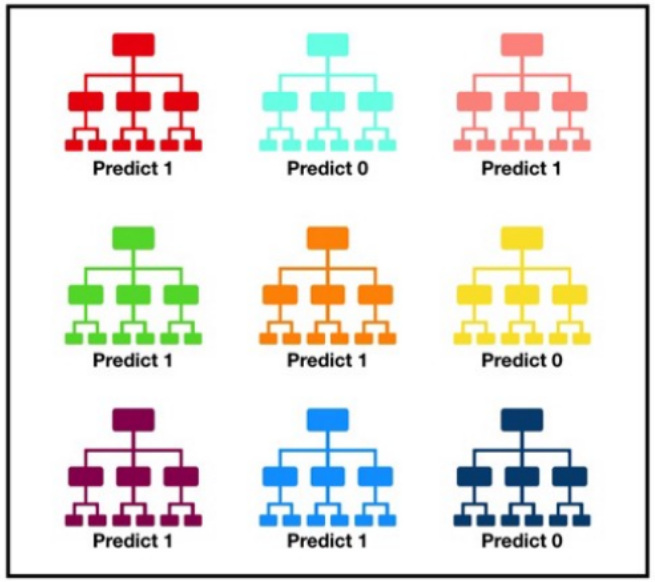
Extension: What is Random Forest

- The random forest is a classification algorithm consisting of many decisions trees.
- It uses bagging and feature randomness when building each individual tree.
- It creates an uncorrelated forest of trees whose prediction by committee is more accurate than that of any individual tree.



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Extension:Why use Random Forest

- A large number of relatively uncorrelated models (trees) operating as a committee will outperform any of the individual constituent models.

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Random Forests

- Random forests provide an improvement over bagged trees by way of a small tweak that decorrelates the trees. This reduces the variance when we average the trees.
- As in bagging, we build a number of decision trees on bootstrapped training samples.
- But when building these decision trees, each time a split in a tree is considered, a random selection of m predictors is chosen as split candidates from the full set of p predictors. The split is allowed to use only one of those m predictors.
- A fresh selection of m predictors is taken at each split, and typically we choose $m \approx \sqrt{p}$ — that is, the number of predictors considered at each split is approximately equal to the square root of the total number of predictors (4 out of the 13 for the Heart data).

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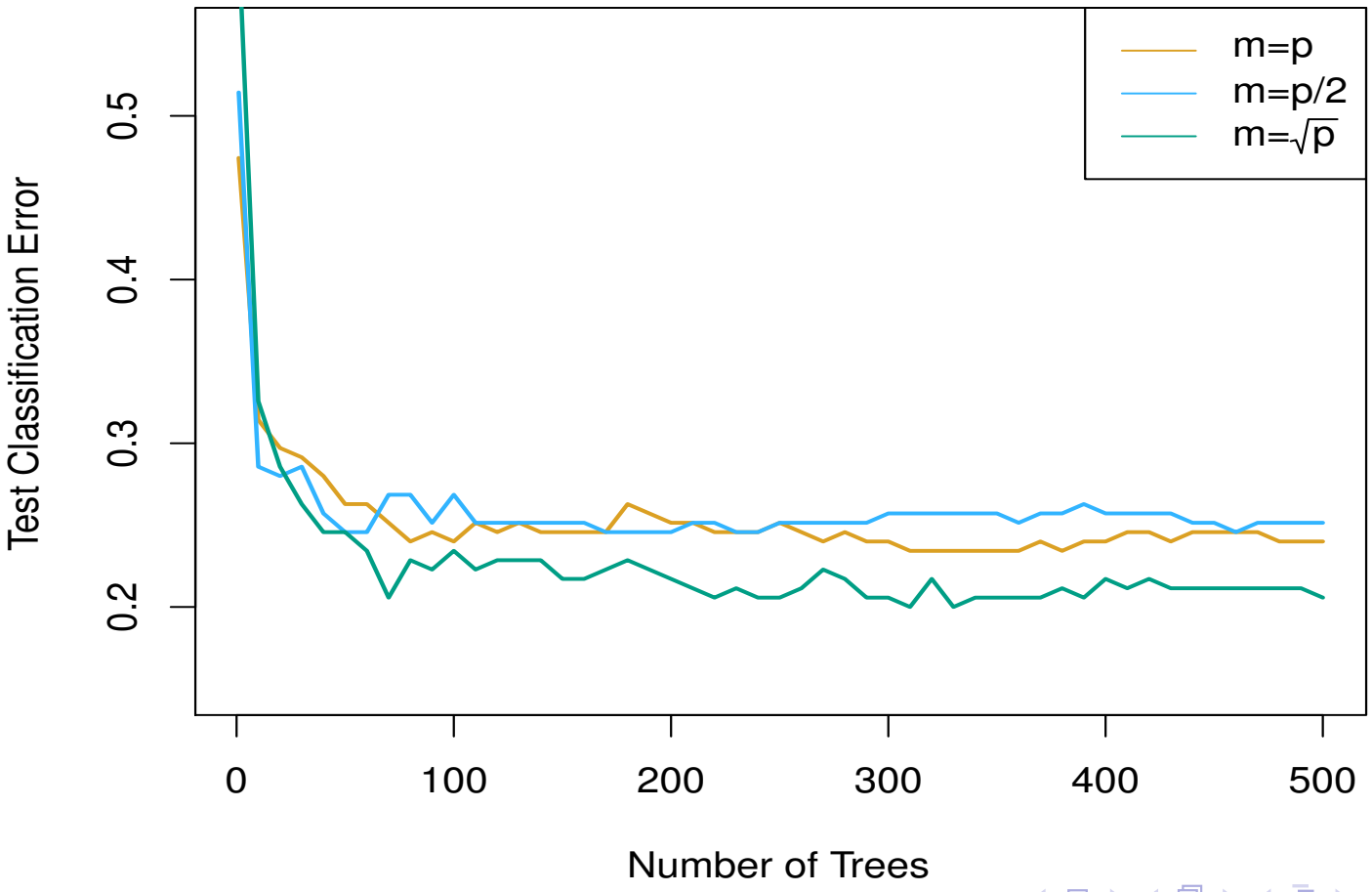
Example: Gene expression data

- We applied random forests to a high-dimensional biological data set consisting of expression measurements of $4,718$ genes measured on tissue samples from 349 patients.
- There are around $20,000$ genes in humans, and individual genes have different levels of activity, or expression, in particular cells, tissues, and biological conditions.
- Each of the patient samples has a qualitative label with 15 different levels: either normal or one of 14 different types of cancer.
- We use random forests to predict cancer type based on the 500 genes that have the largest variance in the training set.
- We randomly divided the observations into a training and a test set, and applied random forests to the training set for three different values of the number of splitting variables m.

Results: Gene expression data



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Details of previous figure

- Results from random forests for the fifteen-class gene expression data set with $p = 500$ predictors.
- The test error is displayed as a function of the number of trees. Each colored line corresponds to a different value of m, the number of predictors available for splitting at each interior tree node.
- Random forests ($m < p$) lead to a slight improvement over bagging ($m = p$). A single classification tree has an error rate of 45.7%.

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Extension: Three famous decision trees in machine learning

* CART decision tree [Breiman, *et al.*, 1984]: Gini index.
* ID3 decision tree [Quinlan, 1986]: Entropy index.
* C4.5 decision tree [Quinlan, 1993]: Gain ratio index.

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Summary and Remark

- Bagging
- Random Forests
- Read textbook Chapter 8 and `Python` code